

Assignment
SQL-Practice Jost - Group by
having & left Join

Table ① Stations

Station-id	Station-name	Zone	Capacity	Opened-year
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Table ② Rides

Ride-id	Station-id	Rider-type	Ride-date	duration-min	distance-km	fare
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Q① Select rider-type, Count(*) as total-rides,
Or
Count(ride-id) as total-fare from rides.
Primary key
group by Select rider-type
having Count(*) > 7
order by total-fare desc;

Q② Select Station-id, Count(*) as ride-count,
Round(Avg(duration-min, 2)) as avg-duration from rides
group by Station-id
having avg-duration > 30
order by avg-duration desc;

Q③ Select month(Ride-date) as ride-month,
Count(*) or Count(Ride-id) as total-rides,
Round(Sum(distance-km)) as total-distance
group by ride-month from rides.

having total-distance or round(sum(distance-km))
> 80
Order by ride-months

Q.4 Select rider_type, max(duration-min)
as longest-ride, min(duration-min)
as shortest-ride, round(avg(duration-min))
as avg-ride from rider.
group by rider_type
having longest-ride >= 50
Order by longest-ride desc;

Q.5 Select station-id, rider_type, count(*)
as ride-count, round(sum(distance-km))
as total-distance from rider.
group by station-id, rider_type
having total distance > 10
Order by station-id;

Q.6 Select station-id,
count(distinct(rider_type)) as
ride-type-count, count(ride-id) as
total rides, round(Avg(duration, 2)) as avg-time
from rider
group by station-id
having count(distinct(rider_type)) = 3
Order by station-id;

Q2) Select s.station-id, s.station-name, s-zone,
 r.ride-id, r.ride-type, r.fare from
 stations as s left join rides as r
 on s.station-id = r.station-id
 where Year(r.ride-date) = 2021 and
 Month(r.ride-date) = 8

Order by s.station-id, r.ride-id;

→ यह group by
 नहीं लागू है
 क्योंकि aggregate function
 use नहीं हुआ।

Q3)

Select s.station-id,
 s.station-name,
 s.zone,

ifnull (Count(r.ride-id), 0) as total-rides,

round(ifnull (Sum(fare), 0), 2) as total-revenue
 from stations s

left join rides as r

on s.station-id = r.station-id

group by s.station-id, s.station-name,
 s.zone

Order by
 total-rides desc,
 s.station-id;

because
 aggregate
 function
 use
 hai

Types of SQL

- ① DOL (Data definition language) - It is used to define, create, modify, alter, truncate, Rename, etc. database structure (schema) of a database. In simple words it is a language which is used to modify structure not on data stored in it.

DDL Commands

- ① Create
- ② Alter
- ③ Drop
- ④ Truncate
- ⑤ Rename

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- ② Alter
- ③ Drop
- ④ Truncate
- ⑤ Rename

- ② DML → Data manipulation language - It is a language used to manipulate data, insert, delete, update data stored in a database table.

Commands

Insert
Delete
Update
Select

Insert
Delete
Update
Select

Insert
delete
Update
Select

- ③ DCL → Data Control language - It is a category of SQL commands used to control access to the database. It determines who can access the database and what actions they are allowed to perform.

① GRANT → Give Permission to access the database

② REVOKE → The revoke is used to remove permission which is previously granted to user.

④ TCL → Transaction Control language → Used to manage transaction in a database
↳ A transaction is a group of one or more SQL statements which is treated as a single unit of work. TCL ensures that changes made to the database are either saved permanently or canceled completely.

↳ Commit

↳ Rollback

↳ Set transaction

↳ Save Point

↳ Commit

↳ Rollback

↳ Save point

↳ Set transaction

⑤ DQL → Data Query language → It is an language or a category of SQL commands used to retrieve data from a data base

↓
Select

Q. ② What are joins? ^{very important - Types}
of joins in SQL - ^{Left join vs Right join}
join difference (in detail) - ^{what is Self}
Join (read Ex of employee and its managers).

① What are joins - A join is a SQL
operation that combines rows
from two or more tables based
on a related column between
them. In simple words - a join glues
the tables back together at query time
but only in the result, not in storage.

Why Joins become important

① They make normalization practical - Splitting
tables is only good idea if you can
easily put them back together.

② They answer questions no single table
can answer.

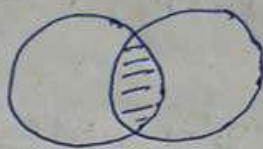
③ They keep data accurate - Because each
fact is stored in one place. Update
credit phone once, and every report
shows the new no. instantly.

④ They are fast.

- ⑤ Combine data from multiple tables.
- ⑥ Save storage space.
- ⑦ Improve database design.

Types of Joins in SQL

- ① Inner Join - Returns only the rows that have a match in both tables. If there is no match, SQL ignores that row.



- ② Left Join (Left outer join) - All rows from the left table.

+ Only the matching row from the right table.

* If there is no matching row in the right table, SQL returns null for the right table's column.



- ③ Right Join (Right outer join) -

+ All rows from the right table.

* Only the matching rows from the left table.

* If there is no matching row in the left table, SQL returns Null for the left table's column.



④ Full outer join → Returns all rows from both tables, whether a match exists or not.

* If a row has a matching value in both tables, it combines them into one row.

* If a row exists only in the left table, that column from the right table contains null.

* If a row exists only in the right table, the column from the left table contains null.

Difference between left Join & Right Join

Basic

Left
Join

Right
Join

turns all
rows from

left table.

Right table.